# BBP

## RASD & DD Presentation

Alessandro Burato, Gabriele Crimella, Davide Faccini

Politecnico di Milano

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# Outline

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   - Overall Description
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   - Architectural Design
   - Implementation, Integration and Test Plan
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# Introduction

- Key goals (G1-G14):
  - **Data Acquisition**: Automated tracking & manual reporting (G10, G11).
  - **Data Quality**: Merging user contributions (G8) & Anomaly detection (G9).
  - **Value**: Path optimality scoring based on road status (G5).

# Phenomena

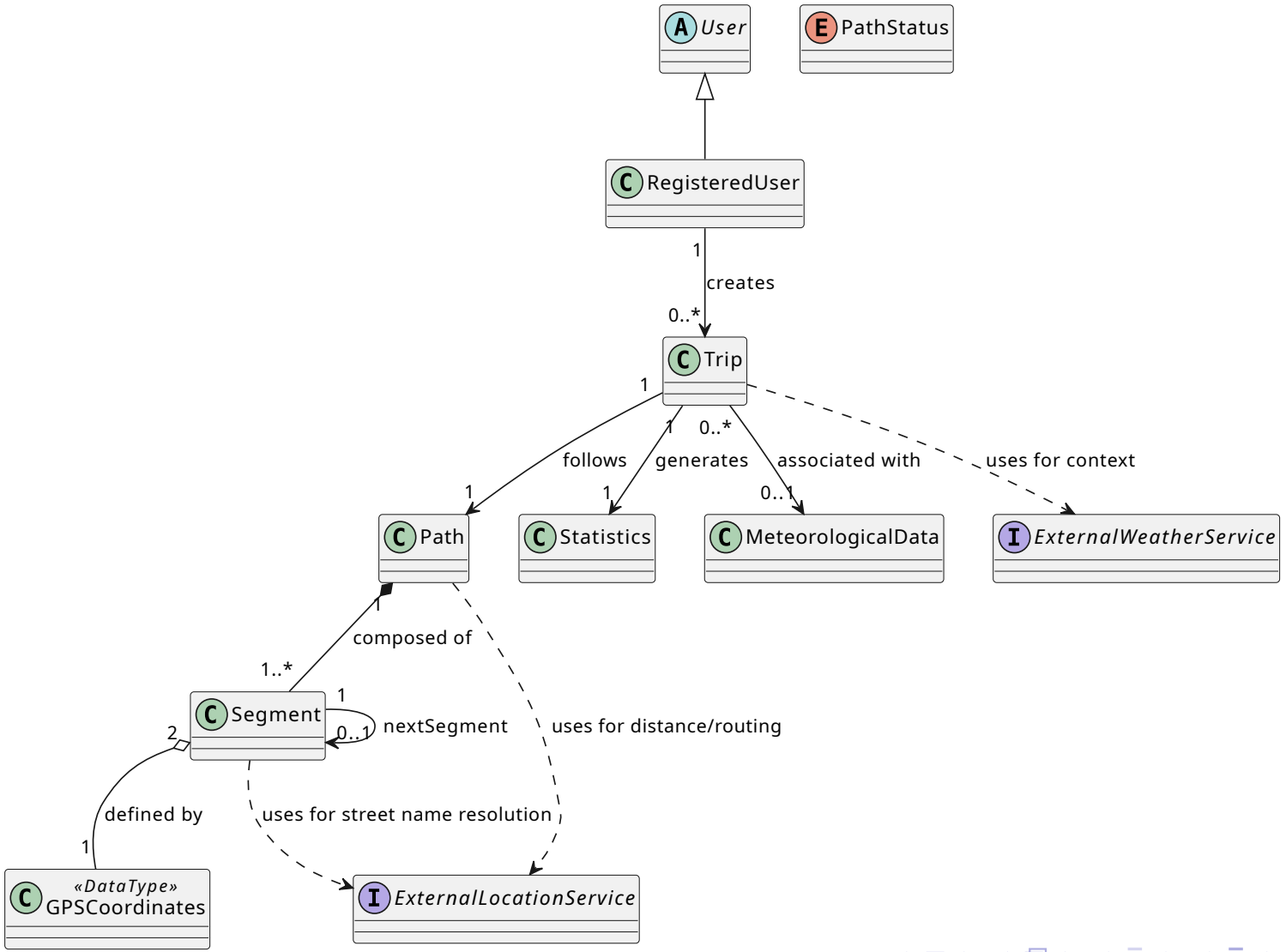
- World Phenomena (WP):
  - *Physical movement*
  - *Road status changes (potholes)*
  - *Weather conditions*
- Shared Phenomena (SP):
  - World-controlled:
    - *Sensors transmitting*
    - *User inputs*
  - Machine-controlled:
    - *Map visualization*
    - *Confirmation requests*

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# Class diagram

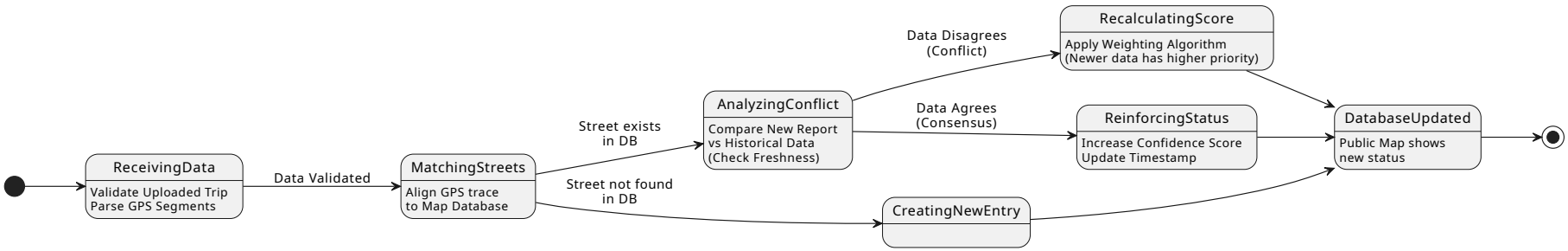


# System Dynamics & Data Lifecycle

- **Acquisition Strategy**: hybrid model using **inference** (speed thresholds) and **topological reconstruction** (manual)
- **Validation Pattern**: sensor anomalies are flagged as "**candidates**" to filter **false positives** via user confirmation
- **Consistency Logic**: server-side **weighting algorithm** resolves conflicts based on data freshness and consensus

# State Diagram Example

## Path Aggregation

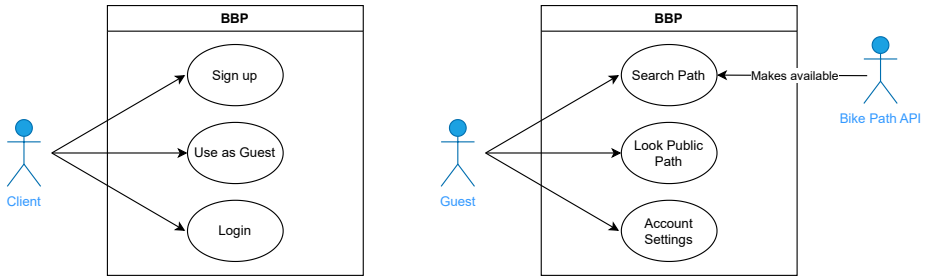


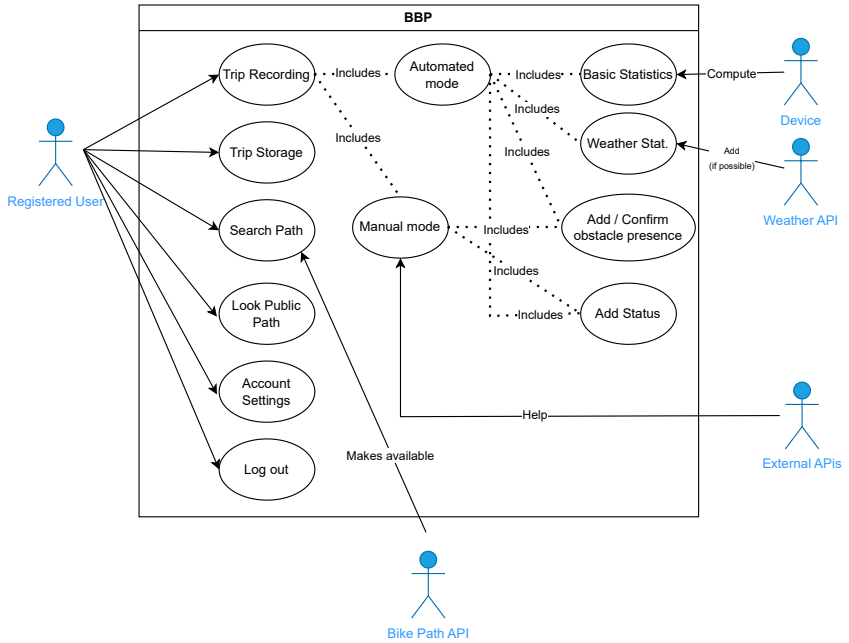
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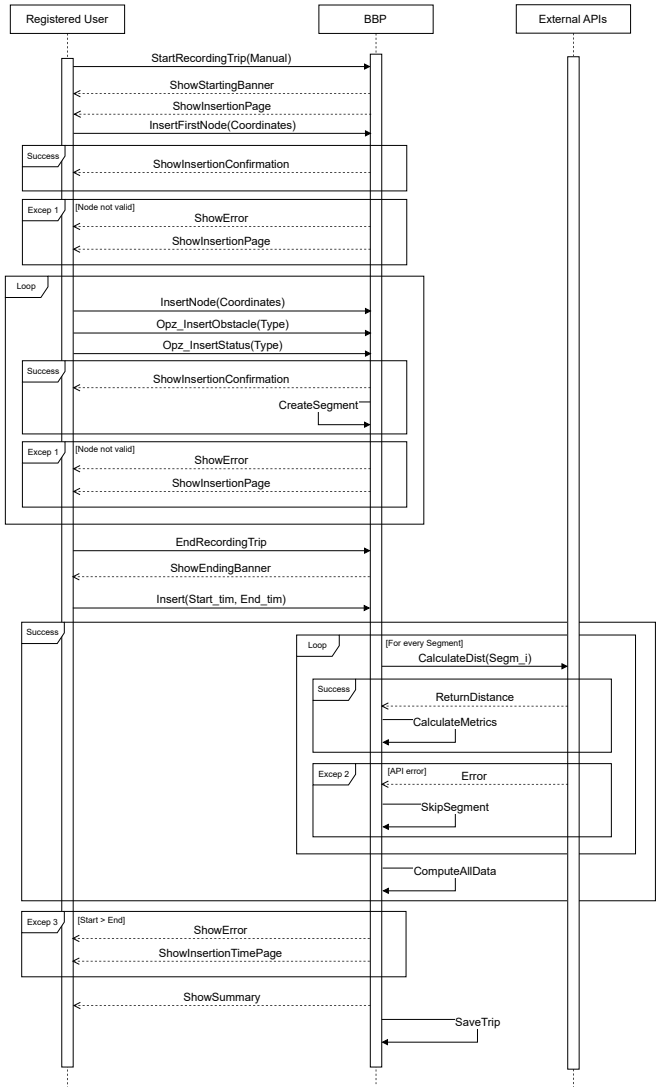
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# Specific Requirements



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# Specific Requirements

- **Realibility**
- **Availability**
- **Security**
- **Maintainability**
- **Portability**

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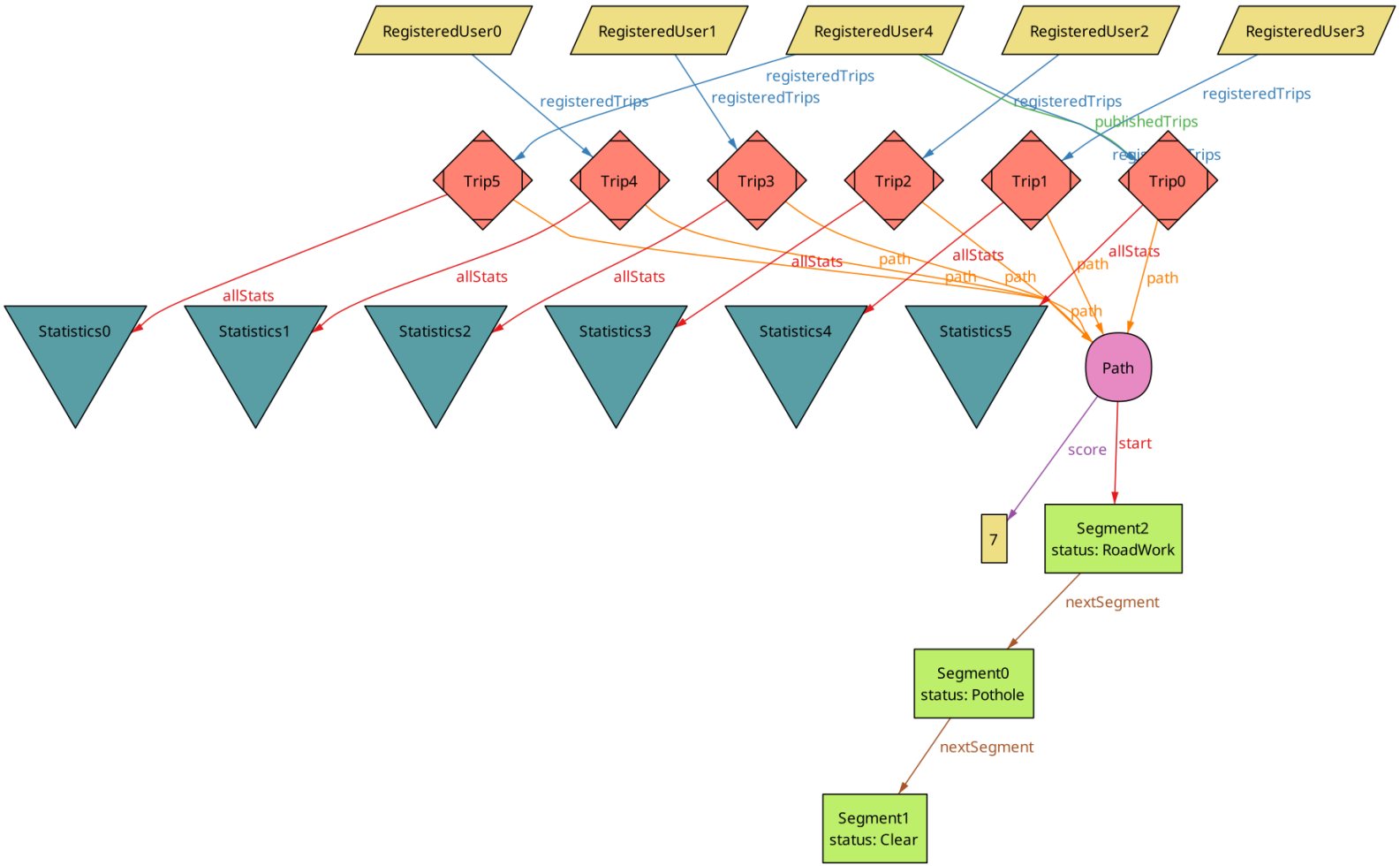
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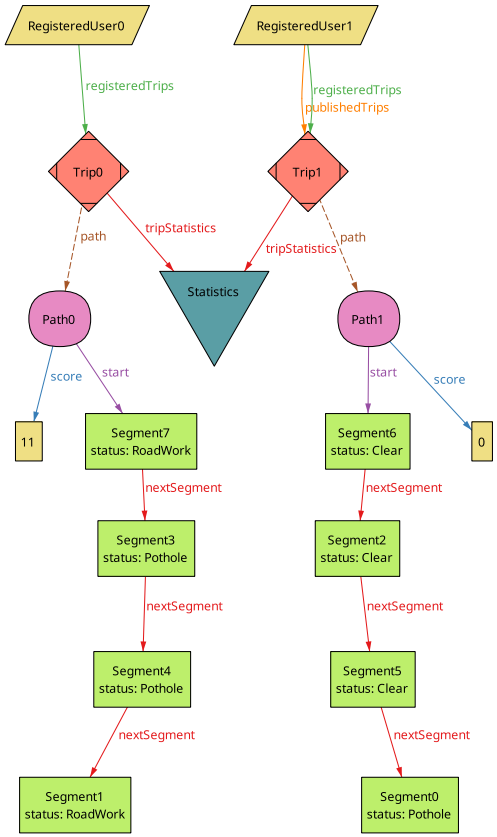
# Alloy

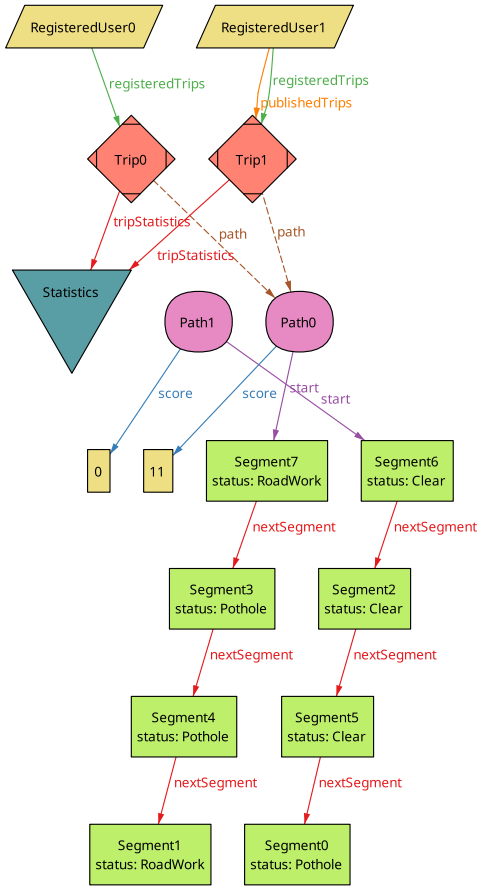
- **Static model analysis**
- **Merging of two paths**

# Static Model Analysis



# Merging of Two Paths



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Architectural Design
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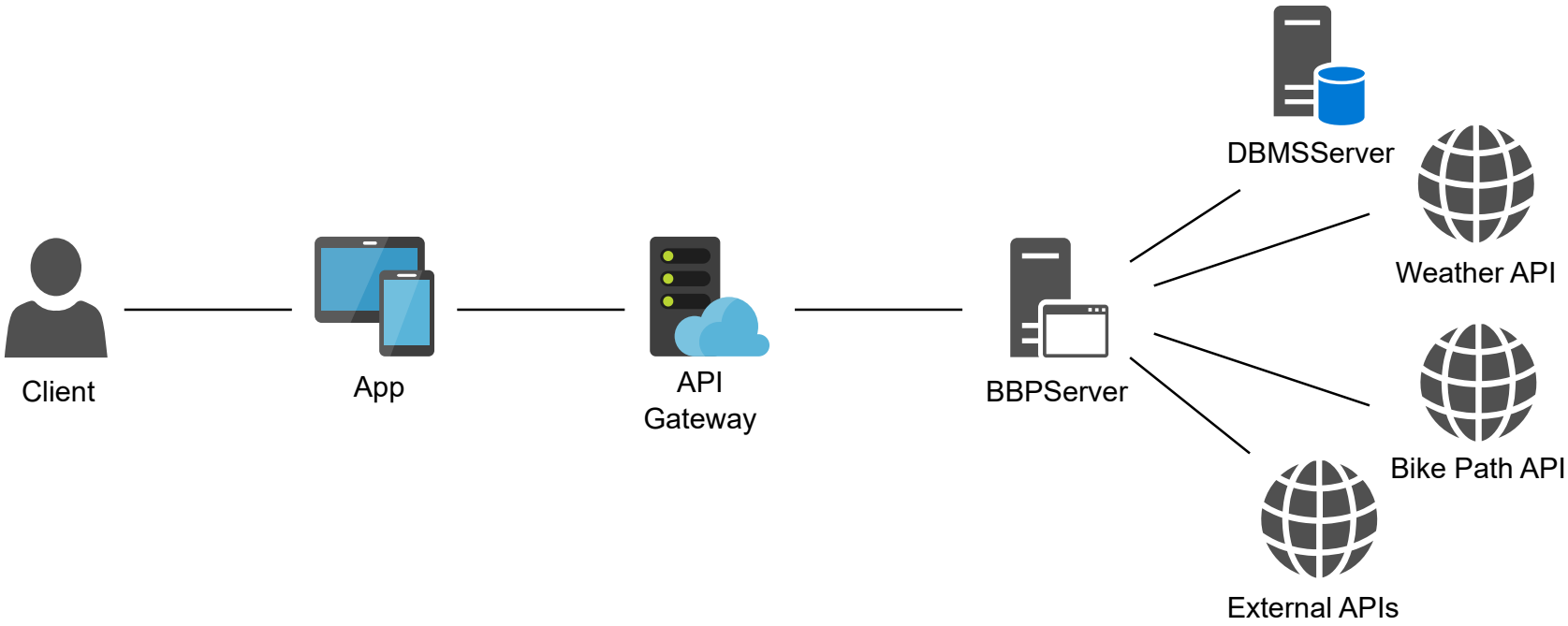
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Architectural Design

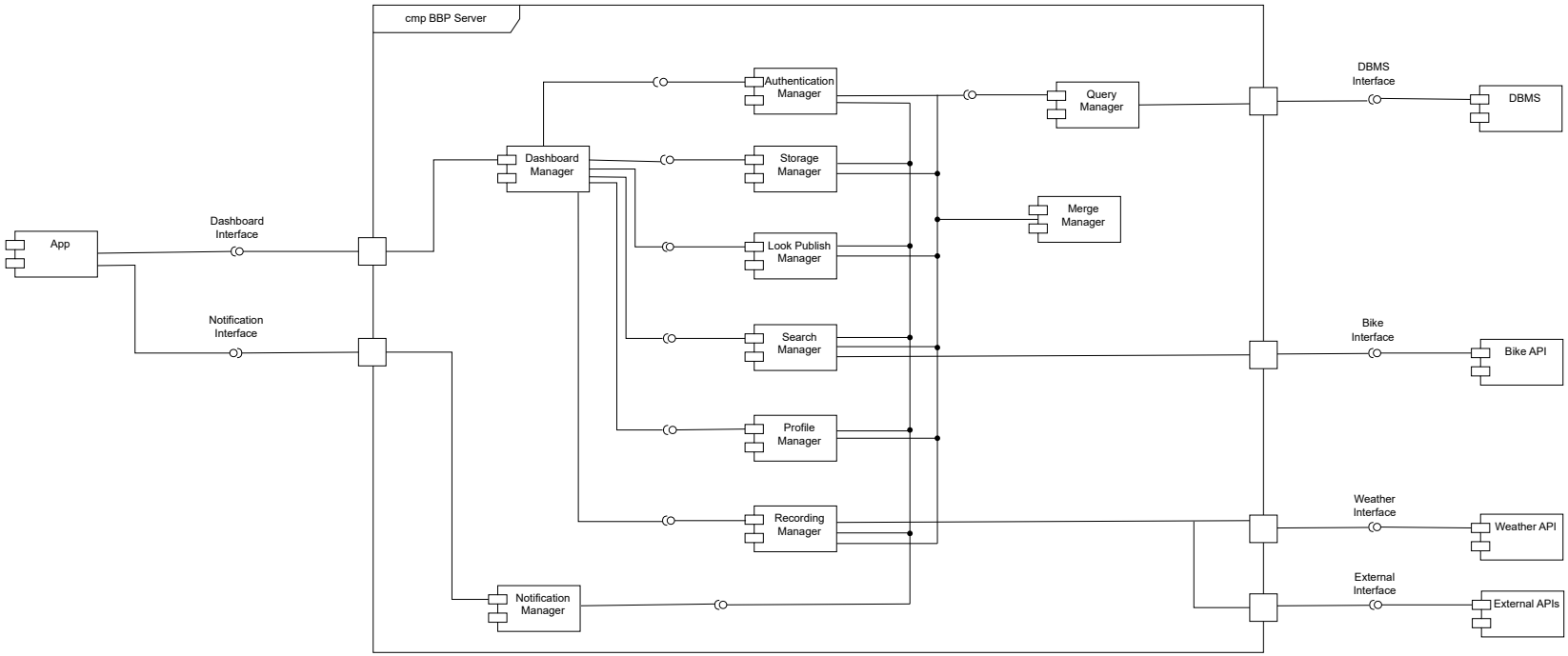
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# Architectural Design



# Architectural Design



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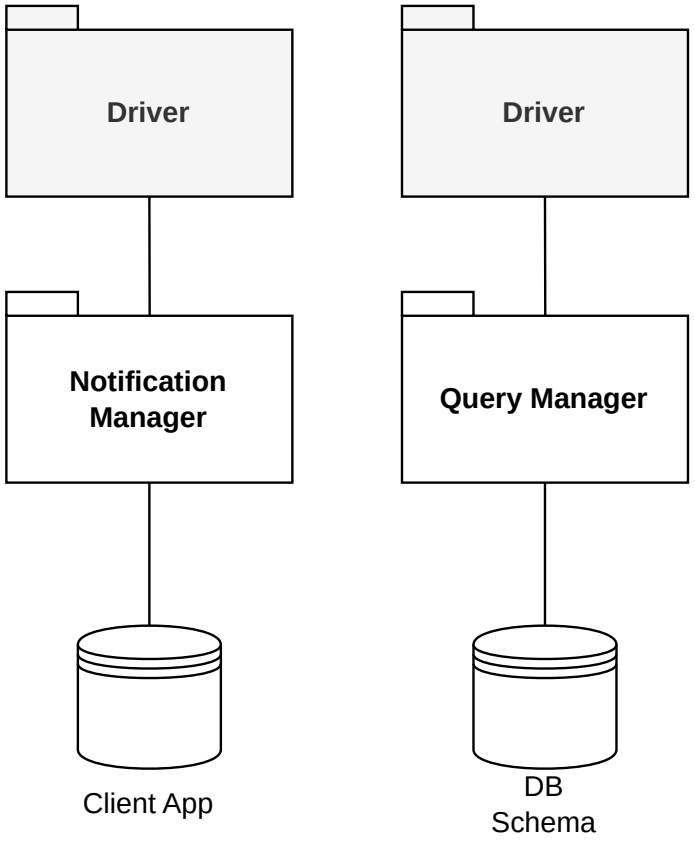
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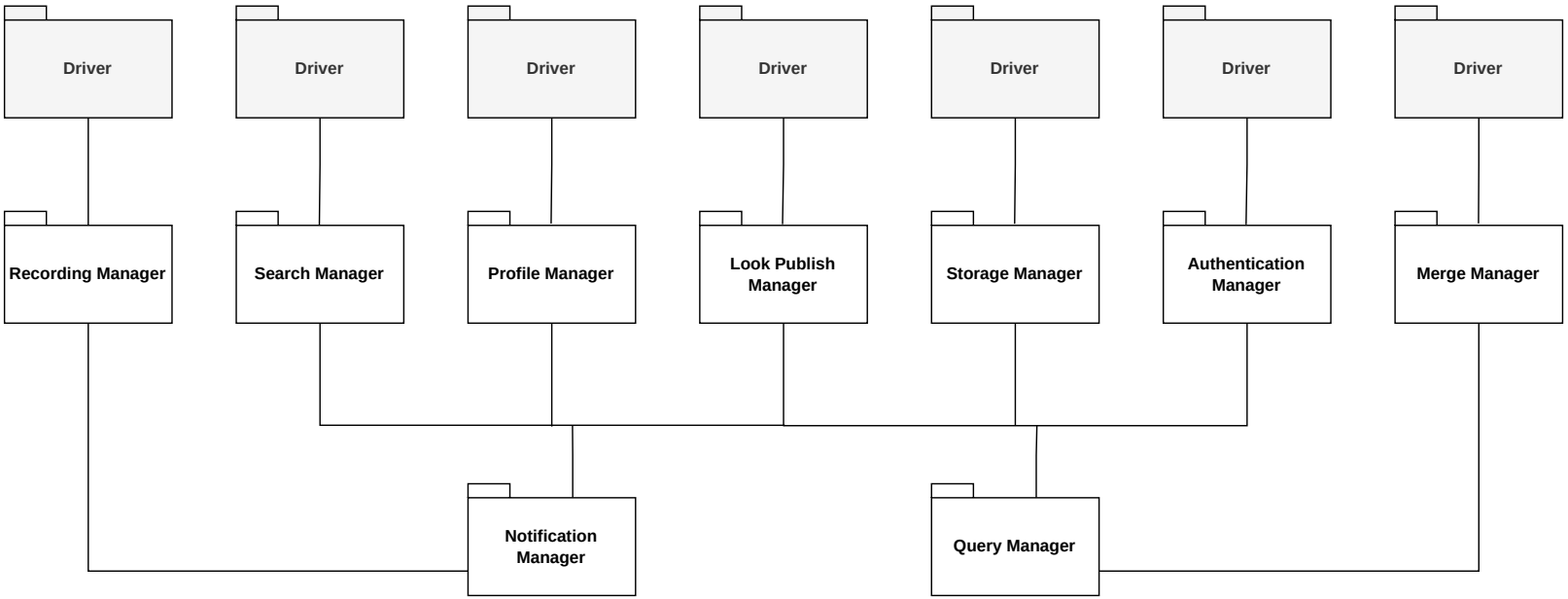
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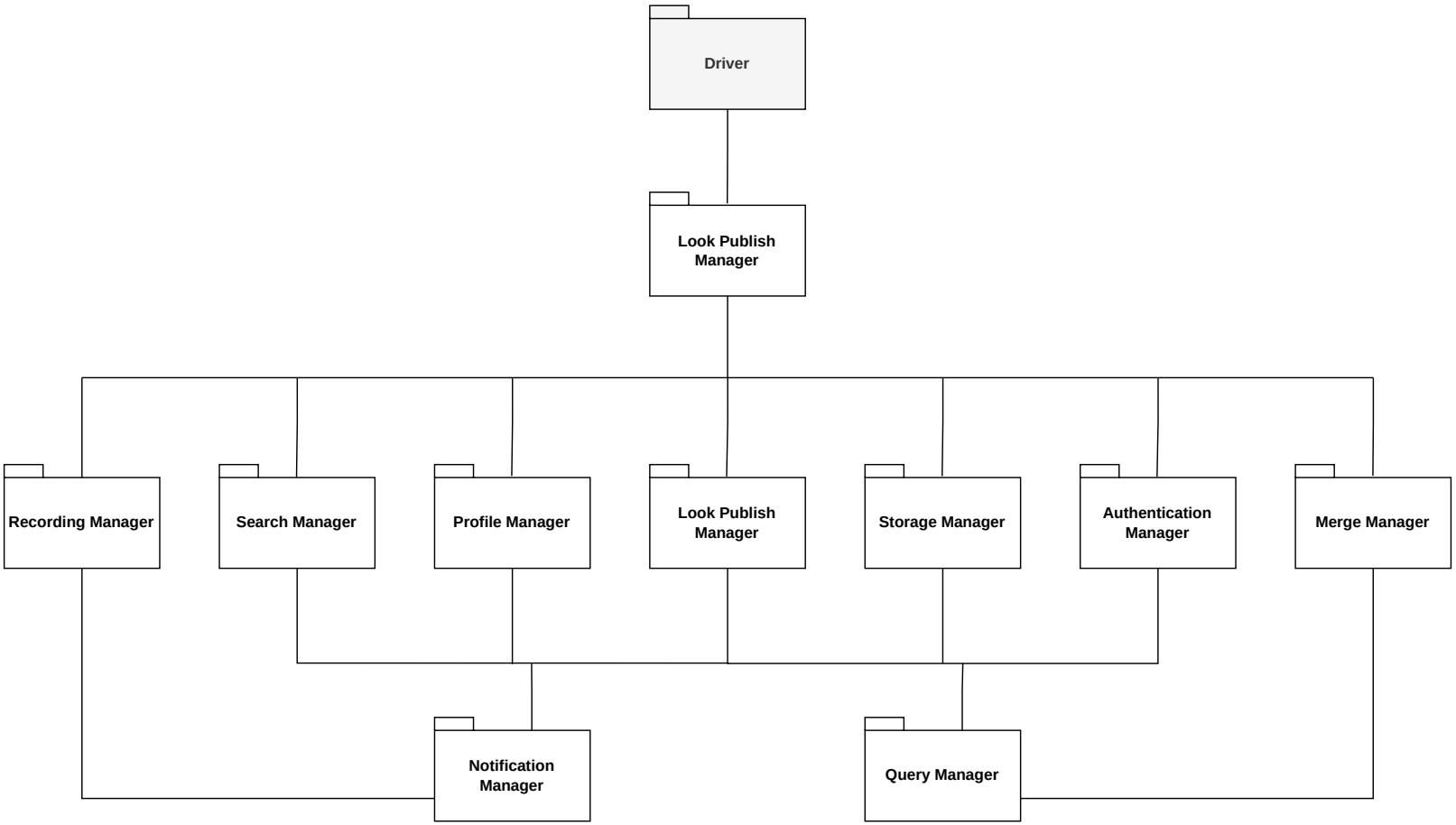
# Query and Notification Manager



# Parallel Integration



# Dashboard Manager



# System Testing Strategy

- **Functional Testing**
- **Load Testing**
- **Performance Testing**
- **Stress Testing**
- **User Interface Testing**

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# References

[1] Author, *Title of the Paper*, Journal, Year.

# Thank you!